

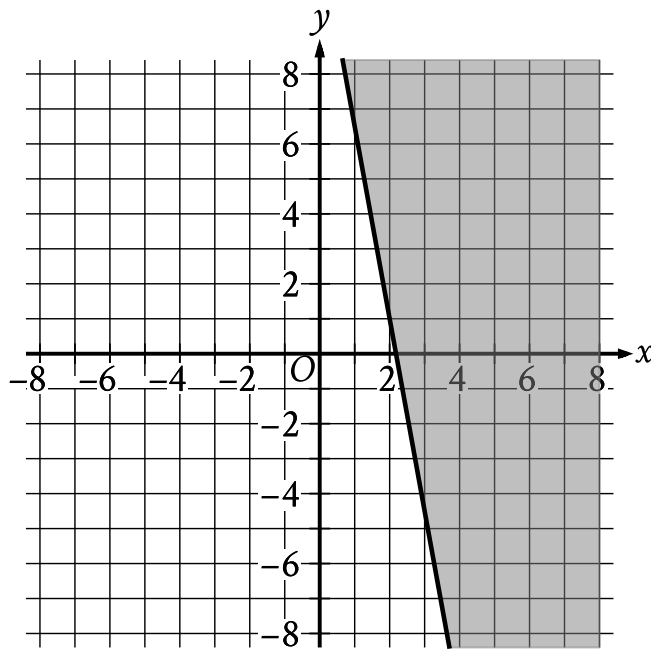
☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

08



- The boundary of the inequality is a solid line.
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (1.0 comma 6.5)
 - (3.0 comma negative 4.5)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents solutions to an inequality. Which ordered pair x, y is a solution to this inequality?

- A. 0, -4
- B. 0, 4
- C. -4, 0

D. 4,0

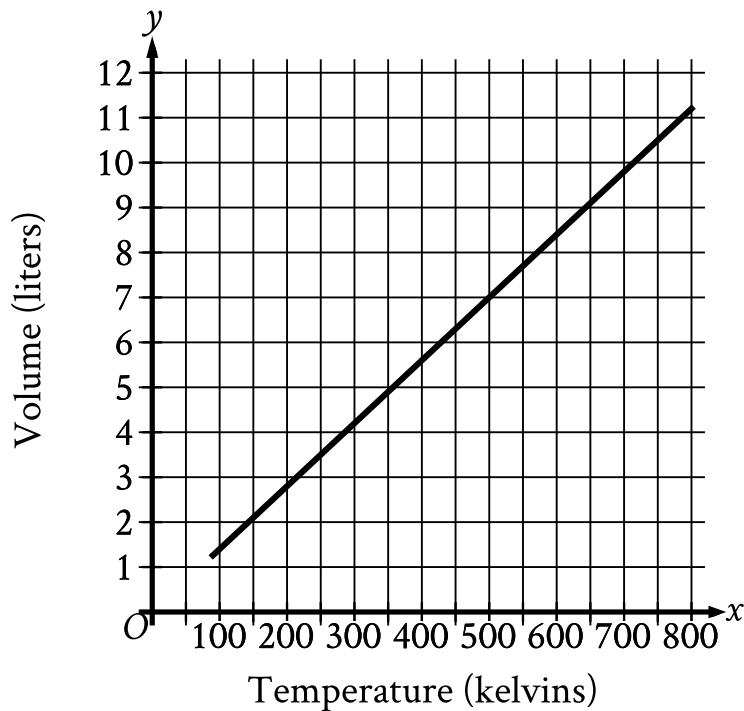
An employee at a restaurant prepares sandwiches and salads. It takes the employee 1.5 minutes to prepare a sandwich and 1.9 minutes to prepare a salad. The employee spends a total of 46.1 minutes preparing x sandwiches and y salads. Which equation represents this situation?

- A. $1.9x + 1.5y = 46.1$
- B. $1.5x + 1.9y = 46.1$
- C. $x + y = 46.1$
- D. $30.7x + 24.3y = 46.1$

What is the solution to the equation $2x + 3 = 7$?

- A. 1
- B. 1.5
- C. 2
- D. 4

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume y , in liters, of the hydrogen when its temperature is x kelvins.



- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.3).
- The line passes through the following approximate points:
 - (90 comma 1.3)
 - (500 comma 7)

What is the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins?

A. 0

B. $\frac{7}{500}$

C. 7

D. $\frac{500}{7}$

$$3x = 12$$

$$-3x + y = -6$$

The solution to the given system of equations is x, y . What is the value of y ?

- A. -3
- B. 6
- C. 18
- D. 30

The point $(8, 2)$ in the xy -plane is a solution to which of the following systems of inequalities?

A. $x > 0$
 $y > 0$

B. $x > 0$
 $y < 0$

C. $x < 0$
 $y > 0$

D. $x < 0$
 $y < 0$

$$x + y = 75$$

The equation above relates the number of minutes, x , Maria spends running each day and the number of minutes, y , she spends biking each day. In the equation, what does the number 75 represent?

- A. The number of minutes spent running each day
- B. The number of minutes spent biking each day
- C. The total number of minutes spent running and biking each day
- D. The number of minutes spent biking for each minute spent running

$$\frac{4x}{5} = 20$$

In the equation above, what is the value of x ?

- A. 25
- B. 24
- C. 16
- D. 15

The function f is defined by $fx = \frac{1}{10}x - 2$. What is the y -intercept of the graph of $y = fx$ in the xy -plane?

A. $-2, 0$

B. $0, -2$

C. $0, \frac{1}{10}$

D. $\frac{1}{10}, 0$

$$4x = 20$$

$$-3x + y = -7$$

The solution to the given system of equations is x, y . What is the value of $x + y$?

- A. -27
- B. -13
- C. 13
- D. 27